

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (IV)	Branch: Computer Science & Engineering
Subject: Discrete Mathematics	Course Code: B022411(014)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives:

1. To introduce a number of discrete mathematical structures found to be serving as tools in the development of theoretical computer science.
2. Course focuses on how discrete structures actually helped computer engineers to solve problems occurred in the development of programming languages.
3. Course highlights the importance of discrete structures towards simulation of a problem in computer science engineering.

UNIT-I MATHEMATICAL LOGIC & BOOLEAN ALGEBRA: Basic concept of mathematical logic, Statements, Connectives, Conditional and biconditional statements, Logical equivalence, Logical implication & quantifiers, Basic concept of Boolean Algebra, Properties of Boolean Algebra, Boolean functions, Disjunctive & conjunctive normal forms of Boolean functions, Applications of Boolean Algebra in switching circuits & logic circuits.

UNIT-II SET THEORY, RELATIONS, FUNCTIONS: Basic concept of set theory, Relations, Properties of relation in a set, Equivalence relation, Composition of relations, Partial order & total order relations, Lattices & Hasse diagram, Introduction to function, Inverse, Identity, Injective, Surjective & Bijective functions, Composition of functions and some special functions.

UNIT-III ALGEBRAIC STRUCTURES: Groups, Subgroups, Cosets, Lagrange's theorem, Isomorphism, Automorphism, Homomorphism, Codes & group codes, Rings, Integral domains and Fields.

UNIT-IV GRAPH THEORY: Introduction to graph theory, Walks, Paths & Circuits, Types of graphs, Shortest path problems, Eulerian and Hamiltonian graphs, Basic concept of tree: spanning tree, minimum spanning tree, search tree, rooted binary tree, Cut sets, Network flow, Matrix representation of graphs.

UNIT-V COMBINATORICS: Permutation and combination, Pigeon-hole principle, Mathematical induction, Principle of Inclusion and Exclusion, Generating function, Recurrence relation.

Text Books:

1. Elements of discrete mathematics by C.L. Liu, Tata McGraw-Hill, publications.
2. Discrete Mathematical structures, by Bernard Kolman, Robert C. Busby and Sharon Cutler Ross, Pearson Education.

Reference Books:

1. A Text Book of Discrete Mathematics, Swapan Kumar Sarkar, S. Chand & Company Ltd.
2. Graph theory with applications to engineering and computer science, by Narsingh Deo, Prentice Hall of India.
3. Discrete mathematics for computer scientists and mathematicians, by J.L. Mott, A. Kandel and T.P. Baker, Prentice Hall of India.
4. Discrete Mathematical Structures with applications to computer science, by J.P. Tremblay and R. Manohar, Tata McGraw-Hill.

Course Outcomes:

After completion of this course students will be –

1. Able to apply mathematical logic and Boolean algebra in switching circuits & logic circuits.
2. Familiar with set theory, relation and functions.
3. Familiar with algebraic structures, graph theory and combinatorics.
4. Able to solve problems in various fields in computer science, specially networking

Printed Pages – 6

Roll No. :

B022411(014)

B. Tech. (Fourth Semester) Examination,

Apri-May 2022

(CSE Branch)

(AICTE Scheme)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

*Note : Attempt all questions. Part (a) is compulsory
from each unit & solve any **two** parts from
(b), (c) & (d) of each unit.*

Unit-I

1. (a) Write short notes on Quantifier. 4
- (b) Define Boolean Algebra prove the following identities : 8

[2]

- (i) $a + a = a$
(ii) $a \cdot a = a$
(iii) $(a')' = a$
- (c) Explain the connectives conjunction, Disjunction, Conditional and Biconditional with truth tables. 8
- (d) Explain normal forms of Boolean functions change the following Boolean function to disjunctive normal form : 8

$$f(x, y, z) = \left[x + (x' + y)' \right] \left[x + (y' \cdot z')' \right]$$

Unit-II

2. (a) Explain floor and ceiling functions with examples. 4
- (b) Define partial order relation. Prove that the relation “ a divides b ”, it there exists an integer c such that $ac = b$ and is denoted by a/b , on the set of all positive integers N is a partial order relation. 8
- (c) Define Injective, Surjective and Bijective function. If Q is the set of all rational numbers and $f : Q \rightarrow Q$

[3]

- is defined by $f(x) = 3x + 2$, $x \in Q$, then prove that f is bijective function. Also find F^{-1} . 8
- (d) Define Lattice. Let L be the set of all factors of 12 and let “ $\mid$ ” be the divisibility relation on L . Show that $(L, \mid)$ is a lattice. 8

Unit-III

3. (a) Define group code. 4
- Show that $(5, 2)$ encoding function $e/B^2 \rightarrow B^5$ defined by
- $$e(00) = 00000 \quad e(01) = 01110$$
- $$e(10) = 10101 \quad e(11) = 11011$$
- is a group code.
- (b) Show that $S = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ for the operations $+$, $\times$ is an integral domain but not a field. 8
- (c) State and prove Lagrange's theorem. 8

[4]

- (d) Define Isomorphism of groups. If G be the multiplicative group of three cube roots of unity, u
 $G = (\{1, w, w^2\}, \cdot)$ where $w^3 = 1$ and G be the
 additive group of integers module 3 u .
 $G' = (\{0, 1, 2\}, +_3)$ then show that G and G' is
 isomorphic.

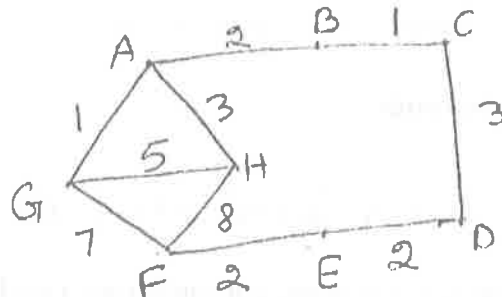
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Unit-IV

4. (a) Explain Walk, Path and Circuit is a graph with
 examples.
 (b) Solve the travelling salesman problem for the
 following weighted graph.

4

8

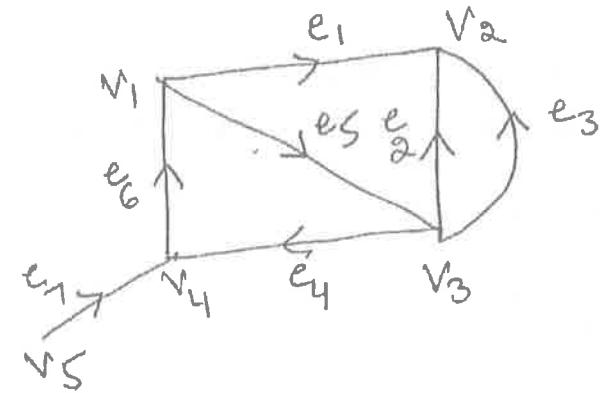


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[5]

- (c) Define and find the incidence and adjacency matrix
 of the following diagram :

8



- (d) What is a simple graph. Show that the maximum
 number of edges in a simple graph with n vertices

$$\text{is } \frac{n(n-1)}{2}.$$

8

Unit-V

5. (a) Determine the generating function for the following
 sequence

4

$$a_r = \begin{cases} 2^r & \text{if } r \text{ is even} \\ -2^r & \text{if } r \text{ is odd} \end{cases}$$

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[6]

(b) Use mathematical induction to show that $n! \geq 2^{n-1}$
for $n \geq 1$. 8

(c) Find the number of integers between 1 and 250 that
are divisible by any of the integers 2, 3, 5 and 7. 8

(d) Solve the difference equation

$$a_r - 4 a_{r-1} + 4 a_{r-2} = (r+1) 2^r \quad 8$$



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**B. Tech. (Fourth Semester) Examination,
April-May 2023**

(New Schemes)

(CSE Branch)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. Part (a) is compulsory & carries equal 4 marks. Solve any two from (b), (c) & (d) of each unit & carries equal 8 marks.

Unit-I

1. (a) Define Logic and their basic operations. 4
- (b) What are logic connectives. Explain using truth tables. 8

- (c) What are disjunctive and conjunctive normal forms convert the function $x + y'$ into disjunctive and conjunctive normal forms of variables x, y, z . 8
- (d) Explain the application of Boolean algebra in switching circuits and logic circuits with examples. 8

Unit-II

2. (a) Explain countable and uncountable sets. 4
- (b) What is an equivalence relation. Explain with examples prove that for two equivalence relations in set their intersection is also an equivalence relation. 8
- (c) What is composition of function. Explain If $f: R \rightarrow R$ is defined as $f(x) = x^2 - 2x - 3$ and the function $g: R \rightarrow R$ is defined as $g(x) = 3x - 4$, find $(g \circ f)(x)$ and $(f \circ g)(x)$. 8
- (d) Define Lattice. Let L be the set of all factors of 12 and let " $\mid$ " be the divisibility relation on L . Show that $(L, \mid)$ is a lattice. 8

Unit-III

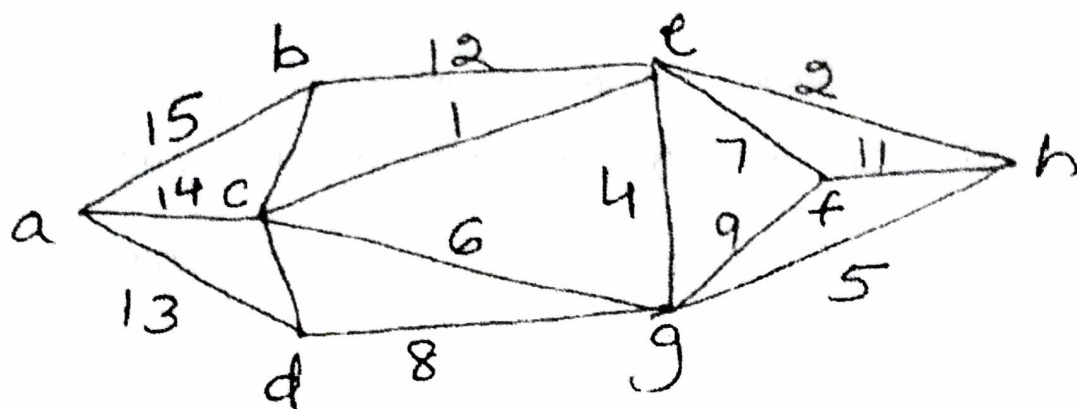
3. (a) Define Group code. 4
- (b) What is homomorphism and isomorphism of groups. Explain with examples. Show that the group $\{z_6, +\}$ is a homomorphic image of the group $\{z, +\}$. 8
- (c) Show that the set of Cube roots of unity is an abelian group with respect to multiplication. 8
- (d) What is a field. For the set $I_4 = \{0, 1, 2, 3\}$. Show that the modulo 4 system is a field. 8

Unit-IV

4. (a) Define Handshaking theorem for graphs. 4
- (b) Explain different operations on graph by taking suitable examples. 8
- (c) Explain the following : 8
- (i) Eulerian and Hamiltonian graph
- (ii) Binary tree and spanning tree

- (d) Explain Kruskal's algorithm to find the minimum spanning tree for the following graph :

8



Unit-V

5. (a) If $nP_2 = 72$, find the value of n .

4

- (b) By mathematical induction prove that

$$(1+x)^n > 1+nx.$$

8

- (c) Solve the fibonacci sequence recurrence relation

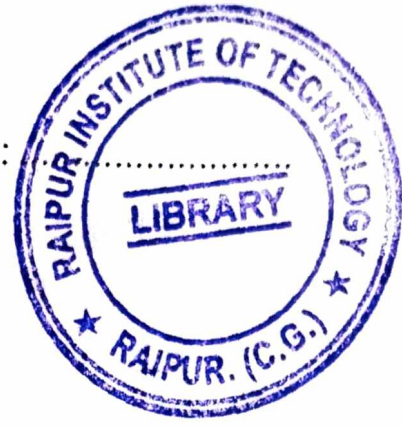
$$f_n = f_{n-1} + f_{n-2}.$$

8

- (d) Use generating function to solve

8

$$a_n = 9a_{n-1} + 20a_{n-2} = 0, \quad a_0 = -3, \quad a_1 = -10.$$



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B. Tech. (Fourth Semester) Examination,

Nov.-Dec. 2023

(CSE Branch)

(New Scheme)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. Part (a) is compulsory & carries equal 4 marks. Solve any two from (b), (c) & (d) of each unit & carries equal 8 marks.

Unit-I

1. (a) Define Boolean Algebra.

4

(b) Prove that :

$$(p \Leftrightarrow q) \wedge (q \Leftrightarrow r) \Rightarrow (p \Leftrightarrow r)$$

is a tautology.

8

- (c) Change the following bookean function to conjunctive normal form

8

$$f(x, y, z, t) = (x' \cdot y + x \cdot y' \cdot z' + x \cdot y' \cdot z + x' \cdot y' \cdot z' \cdot t + t')$$

- (d) Draw the switching circuit of the function

$$F(x, y, z) = x \cdot y' \cdot (z + x) + y \cdot (y' + z)$$

and replace it by simplified one.

8

Unit-II

2. (a) Define Equivalence Relation.

4

- (b) If A and B have n elements in common, how many elements do $A \times B$ and $B \times A$ have in common.

8

- (c) Prove that the relation R in the set of natural numbers N , where the meaning of aRb is

$$a^2 - 4ab + 3b^2 = 0, (a, b) \in N$$

is reflexive but neither symmetric nor transitive.

8

- (d) Show that the composite of mappings obeys associative law.

8

Unit-III

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3. (a) State Lagrange's theorem.

4

- (b) Show that the set of all $(m \times n)$ matrices having their elements as real numbers is an infinite abelian group with matrix addition as composition.

8

- (c) Prove that interection of two subgroups of group G , is a subgroup of G .

8

- (d) Prove that every finite integral domain is a field.

8

Unit-IV

4. (a) Define Hamiltonian path, Hamiltonian circuit and Hamiltonian graph.

4

- (b) Let G be a simple graph with n vertices. If G has k components, then the maximum number of edges

that G can have are $\frac{(n-k)(n-k+1)}{2}$.

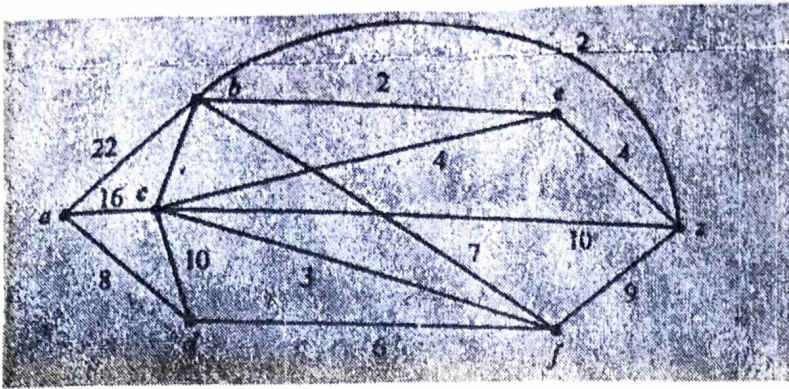
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- (c) Find the shortest path between a and z for the following graph.

8

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- (d) Draw the undirected graph G corresponding to the following adjacency matrix.

8

$$\begin{bmatrix} 1 & 3 & 0 & 1 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 \end{bmatrix}$$

Unit-V

5. (a) Define pigeonhole and generalized pigeonhole principle.

4

- (b) Show that $2^n > n^3$ for all $n \geq 10$ by mathematical induction.

8

- (c) Find the generating function of $a_r = 3^r, r \geq 0$.

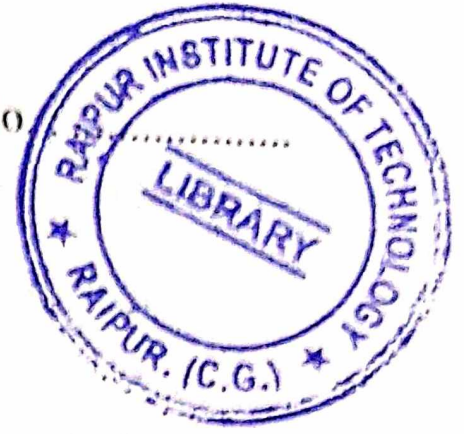
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- (d) Solve the recurrence relation

$$9a_r - 6a_{r-1} + a_{r-2} = 0$$

given that $a_0 = 0$ and $a_1 = 1$.

8



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B. Tech. (Fourth Semester) Examination,

April-May 2024

(New Scheme)

(CSE Branch)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. Part (a) is compulsory & carries equal 4 marks. Solve any two from (b), (c) & (d) of each unit & carries equal 8 marks.

Unit-I

1. (a) Define Tautology.

(b) Define Boolean algebra. Prove the following, for

[2]

each $a, b \in B$, where B is Boolean algebra :

(i) $a + a = a$

(ii) $(a + b)' = a' \cdot b'$

- (c) Define disjunctive normal form and find DNF of the following statement

$$f(x, y, z) = [x + (x' + y)]' \cdot [x + (y' \cdot z')']$$

- (d) Draw the switching circuit of the function

$$f(x, y, z) = xy' \cdot (x + z) + y(z + y')$$

and replace it by a simplified one.

Unit-II

2. (a) Define composition of mappings.

- (b) If A, B, C are any three non-empty sets, then prove that

$$(A - B) \times C = (A \times C) - (B \times C)$$

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[3]

- (c) Define Lattice with an example. Let L be the set of all factors of 12 and let ' $\mid$ ' be the divisibility relation on L . Show that $(L, \mid)$ is a lattice.

- (d) Define inverse function. If $f : X \rightarrow Y$ be a one-one and onto mapping, then prove that $f \circ f^{-1} = I_Y$ and $f^{-1} \circ f = I_X$ where I_X and I_Y are respectively identity mappings of X and Y .

Unit-III

3. (a) In a group $(G, *)$, show that $(a^{-1})^{-1} = a$

- (b) State and prove Lagrange's theorem.

- (c) The necessary and sufficient condition for a non-empty sub-set H of a group $(G, *)$ to be a sub-group is

$$a \in H, b \in H \rightarrow a * b^{-1} \in H$$

where, b^{-1} is the inverse of b in G .

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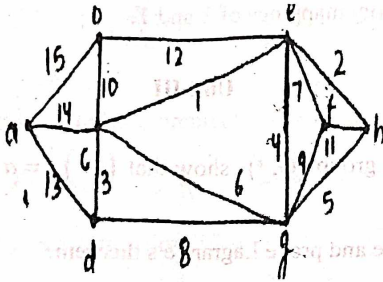
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| 4 |

(d) Prove that every field is an integral domain.

Unit-IV

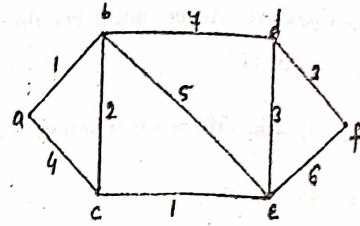
4. (a) Define Eulerian and Hamilton graph.
(b) Define minimum spanning tree. Find the minimum spanning tree of the following graph by Kruskal's algorithm.



- (c) Apply Dijkstra's algorithm to the graph given below, find the shortest path from a to f;

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| 5 |



- (d) Draw the undirected graph represented by the following adjacency matrix A :

$$A = \begin{bmatrix} 1 & 3 & 0 & 1 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 \end{bmatrix}$$

Unit-V

5. (a) Write the Pigeon-hole principle.
(b) Show that

$$1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2}, n \geq 1$$

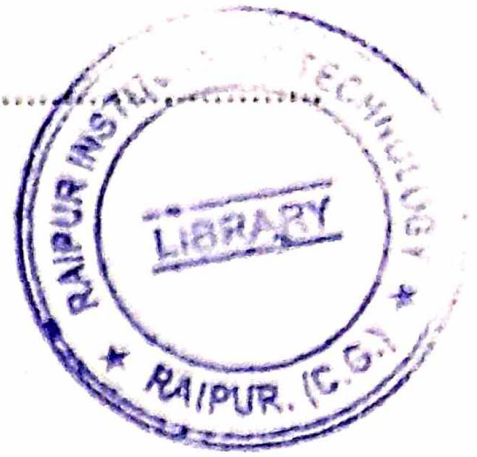
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(c) How many positive integers not exceeding 500 are divisible by 7 or 11?

(d) Solve the following difference equation

$$a_r - 4a_{r-1} + 4a_{r-2} = 2^r$$



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B. Tech. (Fourth Semester) Examination,

Nov.-Dec. 2024

(CSE Branch)

(New Scheme)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. Part (a) is compulsory & carries equal 4 marks. Solve any two from (b), (c) & (d) of each unit & carries equal 8 marks.

Unit-I

1. (a) (i) Write De-Morgan's law for Boolean algebra. 2
- (ii) In a Boolean algebra $(B, +, *)$ prove that : 2

$$a + a'b = a + b$$

[2]

(b) What is Tautologies, show that :

$$(p \Leftrightarrow q \wedge r) \Rightarrow (\sim r \Rightarrow \sim p)$$

are tautology.

8

(c) In the Boolean algebra $(B, +, *)$ express the Boolean

function $F(x, y, z) = x \cdot (y' \cdot z)'$ in its disjunctive

normal form.

8

(d) Draw the switching circuit for switching function

$$F(x, y, z) = x \cdot y \cdot z + (x + y) \cdot (x + z)$$

and replace it by a simpler one.

8

Unit-II

2. (a) If A, B, C are any three sets, then prove that : 4

$$A - (B \cup C) = (A - B) \cap (A - C)$$

(b) If R be a relation in the set of integers z defined by

$$R = \{(x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}, (x - y) \text{ is divisible by } 6\}$$

Then prove that R is an equivalence relation.

8

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[3]

(c) Explain with example :

8

(i) Hasse diagram

(ii) Lattice

(d) (i) Explain Surjective and Bijective function with figure.

4

(ii) If $f: R \rightarrow R$, $f(x) = \cos x$ and $g: R \rightarrow R$,

$g(x) = x^2$ then find $(g \circ f)(x)$ and

$(f \circ g)(x)$ is $f \circ g = g \circ f$?

4

Unit-III

3. (a) Define Ring.

4

(b) Show that the set of all positive rational numbers forms an abelian group under composition defined

$$\text{by } a * b = \frac{ab}{2}.$$

8

(c) Define Homomorphism and Isomorphism of Groups.

8

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PTO

[4]

- (d) For the set $I_4 = \{0, 1, 2, 3\}$, show that the modulo 4 system is a field. 8

Unit-IV

4. (a) Write and explain types of graph. 4
- (b) A simple graph with n vertices and k components cannot have more than $\frac{(n-k)(n-k+1)}{2}$ edges. 8
- (c) Draw the undirected graph represented by adjacency matrix A given by 8

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

- (d) (i) Explain Eulerian graph with it's properties. 4
- (ii) Define Path and Circuit. 4

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[5]

Unit-V

5. (a) (i) Define pigeon hole principle. 2
- (ii) If 9 books are to be kept in 4 shelves these must be at least one Shelf which contains at least 3 books. 2

- (b) Show that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}, \quad n \geq 1$$

by Mathematical induction. 8

- (c) Solve the recurrence relation $9a_r - 6a_{r-1} + a_{r-2} = 0$ given that $a_0 = 0$ and $a_1 = 1$. 8

- (d) Find the sequence $\{y_r\}$ having the generating function G given by 8

$$G(x) = \frac{3}{1-x} + \frac{1}{1-2x}$$

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B. Tech. (Fourth Semester) Examination,

April-May 2025

(New Scheme)

(CSE Branch)

(Discrete Mathematics)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. Part (a) is compulsory & carries equal 4 marks each. Solve any two from (b), (c) & (d) of each unit & carries equal 8 marks each.

Unit-I

1. (a) Define Tautology and Contradiction with example. 4

(b) Establish the equivalence relation

$$p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$$

(c) Define Conjunctive Normal Form (CNF) in the

Boolean algebra $(B, +, \cdot, ')$ and express the

Boolean function $f(x, y, z) = (x \cdot y' + x \cdot z)' + z'$

in its conjunctive normal form.

(d) Draw the switching circuit of the following function

$$f(x, y, z) = x \cdot y + [z \cdot (x' + y')]$$

and replace them by simpler ones, also draw the simpler circuit.

Unit-II

2. (a) Explain the following with examples :

(i) Equivalence Relation

(ii) Partial Order Relation

(b) If U is a Universal set and A and B are its two

subsets, then prove the following De-Morgan's Law

$$(i) \quad (A \cup B)' = A' \cap B'$$

$$(ii) \quad (A \cap B)' = A' \cup B'$$

(c) If R and S be equivalence relations in the set X , then prove $R \cap S$ is an equivalence relation in X . 8

(d) If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be one-one onto mapping, then prove that the mapping $g \circ f: X \rightarrow Z$ is also one-one onto and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}. \quad 8$$

Unit-III

3. (a) Define the following with example : 4

(i) Abelian group and

(ii) Subgroup

(b) Show that the set of cube roots of unity is an abelian group with respect to multiplication. 8

(c) State and prove Lagrange's theorem. 8

(d) Define isomorphism. Let R be the additive group of all real numbers and R_+ be the multiplicative group

of positive real numbers, then show that the following
is isomorphism $f: R \rightarrow R_+$ defined by

8

$$f(x) = e^x \quad \forall x \in R.$$

Unit-IV

4. (a) Explain the following with example.

4

- (i) Incidence Matrix and
- (ii) Adjacency Matrix

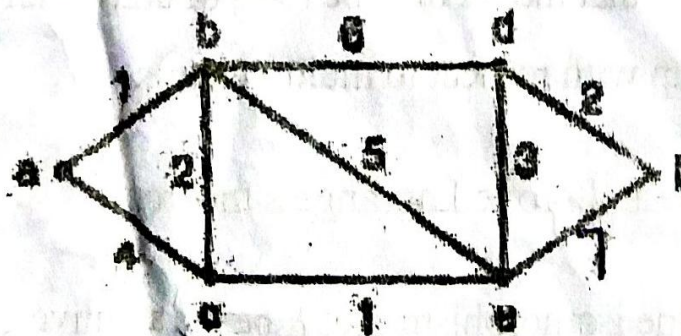
(b) Show that the maximum number of edges in a simple

graph with n vertices is $\frac{n(n-1)}{2}$.

8

(c) Apply Dijkstra's algorithm to the graph given below
and find the shortest path from point a to f .

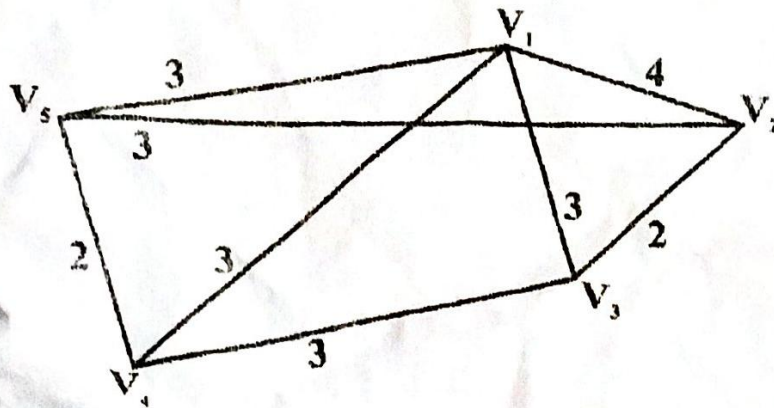
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[5]

- (d) Apply Prim's algorithm to obtain a minimal spanning tree for the following graph :

8



Unit-V

5. (a) Find n , if $2 {}^n P_2 + 50 = {}^{2n} P_2$

4

- (b) Define Pigeonhole principle. How many peoples must you have guarantee that at least 9 of them will have birthday in the same day of the week.

8

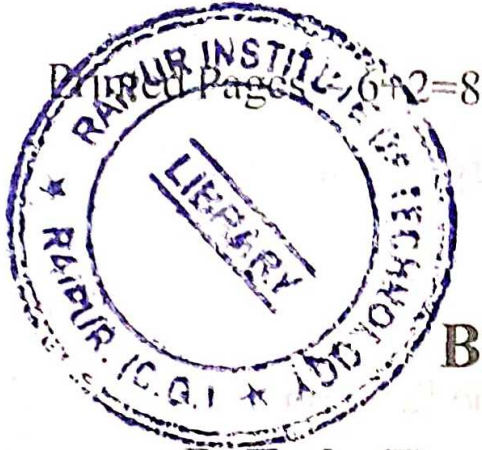
- (c) Find how many integers between 1 and 60 that are not divisible by 2 nor by 3 and nor by 5.

8

- (d) Solve the recurrence relation

8

$$a_{n+2} - 5a_{n+1} + 6a_n = 2, \text{ with } a_0 = 1 \text{ and } a_1 = -1.$$



Roll No. :



B109415(022)

**B. Tech. (Fourth Semester) Examination,
Nov.-Dec. 2025**

**(Computer Science & Engineering, Artificial
Intelligence & Machine Learning)**

DISCRETE STRUCTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all units. part (a) is compulsory and carries equal 4 marks. Solve any two from (b), (c) and (d) of each unit and carries equal 8 marks.

Unit-I

1. (a) Define Tautology and Contradiction. 4
- (b) Prove that : 8
- (i) Involution law $(a')' = a$

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[2]

(ii) $(a.b)' = a' + b'$ for every 'a' and 'b' of a Boolean algebra B.

(c) Draw the switching circuit of the function

$$F(p, q, r) = (p + q)(q + r)(r' + p')$$

and replace it by simple one. Also verify the equivalent circuit by truth table.

8

(d) (i) Define Disjunctive Normal Form and Conjunctive Normal Form.

8

(ii) Change the function $(x.y' + x.z) + x'$ into Disjunctive Normal Form.

8

Unit-2

2. (a) Define Equivalence Relation.

4

(b) If A, B, C are any three sets, then prove that

(i) $A - (B \cup C) = (A - B) \cap (A - C)$

(ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

8

(c) (i) Define Composition of Mapping.

(ii) If $f: R \rightarrow R$ is defined by $f(x) = \cos x$ and

[3]

$g: R \rightarrow R$ is defined by $g(x) = x^3$, then find $(f \circ g)x$ and $(g \circ f)x$.

8

(d) Explain :

8

(i) Injective Function

(ii) Bijective Function

(iii) Partial Order Relation

(iv) Hasse Diagram

(v) Lattice

Unit-3

3. (a) Define :

4

(i) Coset

(ii) Homomorphism of Groups

(b) Let Q_+ be the set of all positive rational number and $*$ is a binary operation on Q_+ defined as

$$a * b = ab/3, a, b \in Q_+$$

Show that $(Q_+, *)$ is a group. Determine identity element and inverse of 'a' in Q_+ .

8

(c) State and proof Lagrange's theorem.

8

(d) Prove that every Field is an Integral Domain.

8



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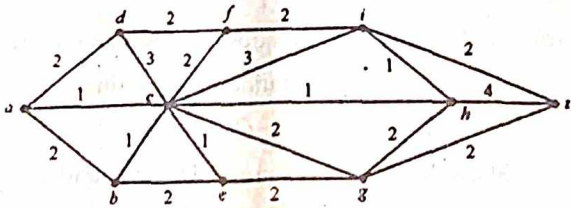
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Unit-4

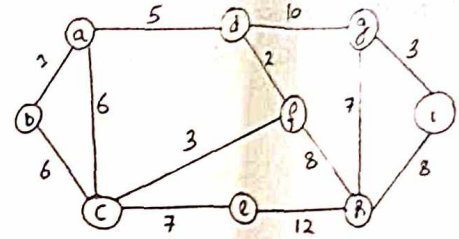
4. (a) Define Hamiltonian path, Hamiltonian circuit and Hamiltonian graph. 4
- (b) Define the following graph : 8
- Undirected and Directed graph
 - Complete graph
 - Isomorphic graph
 - Path and Circuits
 - Cut sets
- (c) Find the shortest path between a and z for the graph. 8



- (d) Find a minimum spanning tree for the graph shown in figure using Prim's Algorithm. 8

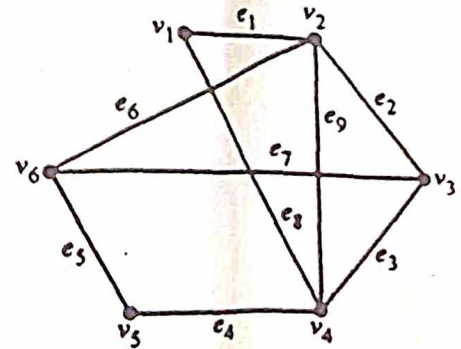


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Unit-5

5. (a) Define Pigeonhole Principal. 4
- (b) (i) Define Adjacency Matrix and Incidence Matrix.
- (ii) Determine the Adjacency Matrix and Incidence Matrix by the picture below : 8



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(c) Solve the difference equation

$$a_r - 4a_{(r-1)} + 4a_{(r-2)} = 2^r. \quad 8$$

(d) Prove that :

$$n! \geq 2^{n-1}, n = 1, 2, 3, \dots \dots \dots 8$$

B022411(014)

**B. Tech. (Fourth Semester) Examination,
April-May 2021**

(AICTE Scheme)

(CSE Engg. Branch)

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Marks : 35

Note : Part (a) of each question is compulsory and carries 4 marks each. All questions are required to be answered, selecting any two from (b), (c) and (d) and carries 8 marks each.

1. (a) Construct converse, inverse and contrapositive of the direct statement :

“If $4x - 2 = 10$ then $x = 3$.”

- (b) Test the validity of the argument :

[2]

"If 8 is even then 2 does not divide 9. Either 7 is not prime or 2 divides 9. But 7 is prime therefore 8 is odd."

- (c) State and prove De-Morgan's law in a Boolean algebra $(B, +, \cdot, ')$.
- (d) Change the following boolean function to disjunctive normal form and conjunctive normal form :

$$f(x, y, z) = (x + y + z)(xy + x'z)'$$

2. (a) Let $A = \{1, 2, 4\}$, $B = \{2, 5, 7\}$ and $C = \{1, 3, 7\}$, then show that

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

- (b) If I is the set of integers and the relation

$$xRy \Leftrightarrow x - y \text{ is an even integer'}$$

then prove that R is an equivalence relation, where $x, y \in I$.

- (c) Show that the inclusion relation ' $\subseteq$ ' is a partial

[3]

ordering on the power set $P(s)$ and draw the Hasse

diagram for the partial ordering $\{(A, B) | A \subseteq B\}$.

- (d) If $f : X \rightarrow Y$ be a one-one and onto mapping then prove that

$$f \circ f^{-1} = I_Y \text{ and } f^{-1} \circ f = I_X$$

where I_X and I_Y are identity mappings of X and Y respectively.

3. (a) Prove that every cyclic group is an abelian group.

- (b) Let Q_+ be the set of all positive rational numbers

and $*$ is a binary operation on Q_+ defined as

$$a * b = \frac{ab}{3}, \forall a, b \in Q_+$$

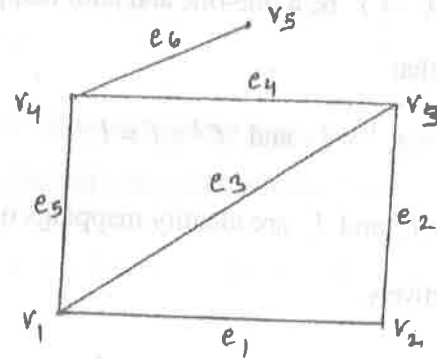
show that $(Q_+, *)$ is a group.

- (c) State and prove lagrange's theorem.

[4]

(d) Prove that every field is an integral domain.

4. (a) Define incidence matrix and find the incidence matrix of the following graph :

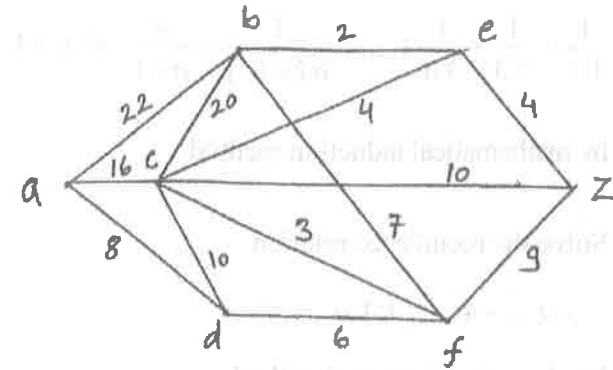


- (b) (i) Prove that the sum of the degrees of odd vertices in a graph is always an even number.

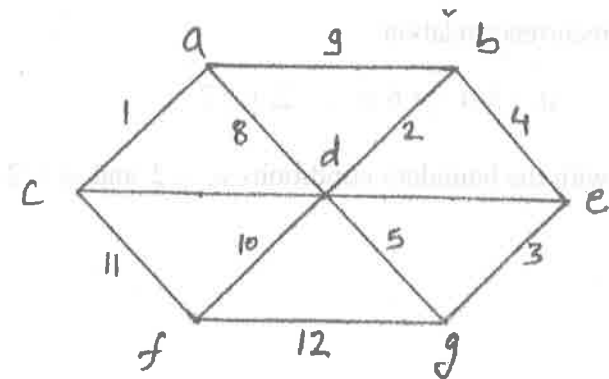
- (ii) Prove that the maximum number of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$.

- (c) Find the shortest path between a and z for the graph given below, where the numbers associated with the edge are the distances between vertices :

[5]



- (d) Find a minimal spanning tree of the graph :



5. (a) State generalized pigeonhole principle. Show that if seven colours are used to paint 50 cars, atleast eight cars will have the same colour.

- (b) Show that :

[6]

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n.(n+1)} = \frac{n}{n+1}, \quad \forall n \geq 1$$

by mathematical induction method.

(c) Solve the recurrence relation

$$a_{r+2} - 4 a_{r+1} + 4 a_r = r^2$$

by characteristics root method.

(d) Solve by the method of generating functions, the recurrence relation

$$a_r - 5 a_{r-1} + 6 a_{r-2} = 2, \quad r \geq 2$$

with the boundary conditions $a_0 = 2$ and $a_1 = 2$.

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Roll No. :

B022411(014)

B. Tech. (Fourth Semester) Examination,

Nov.-Dec. 2021

(CSE Branch) AICTE

DISCRETE MATHEMATICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

***Note : Attempt all questions. Part (a) is compulsory
& solve any two from (b), (c) & (d) of each
unit.***

Unit-I

1. (a) Define Universal and Existential quantifier. 4

(b) Prove that $(p \leftrightarrow q) \wedge (q \leftrightarrow r) \rightarrow (p \leftrightarrow r)$ is a
tautology. 8

[2]

- (c) Obtain the conjunctive normal form of the following function :

8

$$f(x, y, z) = xy' + xz + xy$$

- (d) Prove that for every elements a and b of a Boolean algebra :

8

(i) $(a + b)' = a' \cdot b'$

(ii) $(a \cdot b)' = a' + b'$

Unit-II

2. (a) Define Equivalence Relation.

4

- (b) If A, B, C are any three non empty sets then prove that :

8

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

- (c) Show that the function $f: R \rightarrow R$ defined by $f(x) = 5x^3 - 1$ is one-one onto. Where R is the set of real numbers.

8

[3]

- (d) Explain :

8

(i) Partial Order Relation

(ii) Hasse Diagram

(iii) Lattice

Unit-III

3. (a) Define Algebraic Structure.

4

- (b) Show that the set of all integers I forms a group with respect to binary operation '*' defined by the rule

8

$$a * b = a + b + 1 \quad \forall a, b \in I$$

- (c) Define the following with example :

8

(i) Rings

(ii) Integral Domain

- (d) Prove that the order of each subgroup of a finite group is a divisor of the order of the group.

8

Unit-IV

4. (a) Define the following terms :

4

[4]

(i) Directed Graph

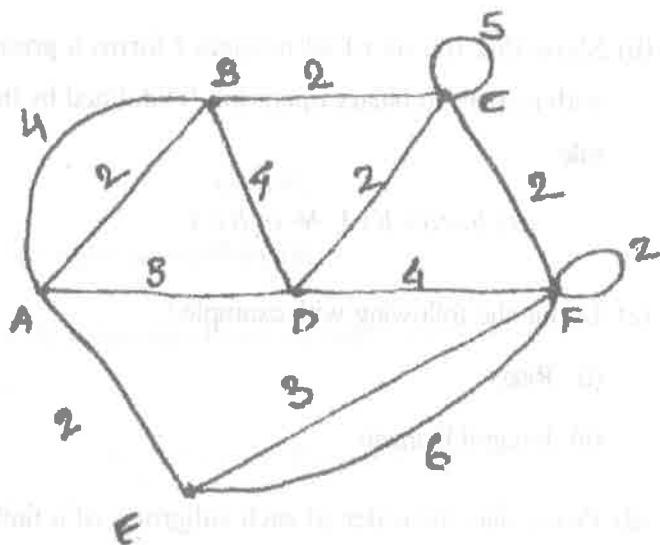
(ii) Undirected Graph

(b) Prove that if G is a tree with n vertices then it has exactly $(n - 1)$ edges.

8

(c) Using Dijkstra's algorithm, find the shortest path between A and F for the following graph.

8

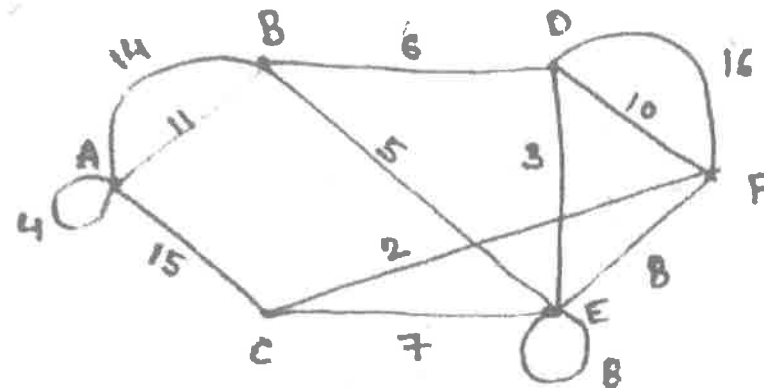


(d) Find the minimal spanning tree of the weighted graph using prim's algorithm.

8

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[5]



Unit-V

5. (a) If ${}^nP_4 = 12 \times {}^nP_2$ then find the value of n .

4

(b) Use method of induction to prove that :

8

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

(c) Find the generating function for

8

$$a_r = 3^r, \quad r \geq 0$$

(d) Solve the recurrence relation

$$9a_r - 6a_{r-1} + a_{r-2} = 0$$

given that $a_0 = 0$ and $a_1 = 1$.

8

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